



Bahria University
Discovering Knowledge
Karachi Campus

DEPARTMENT OF ELECTRICAL ENGINEERING

ENGINEERING DRAWING & CAD (LAB)

Bracket Fan

Submitted By

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Submitted To

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INTRODUCTION:

In this project, a wall-mounted fan was designed and modeled to demonstrate the application of 3D design principles in SolidWorks. The primary objective was to create a detailed and functional representation of a fan that aligns with real-world engineering and manufacturing standards. This report covers the design process, tools used, the working mechanism, and the final outcomes of the project.

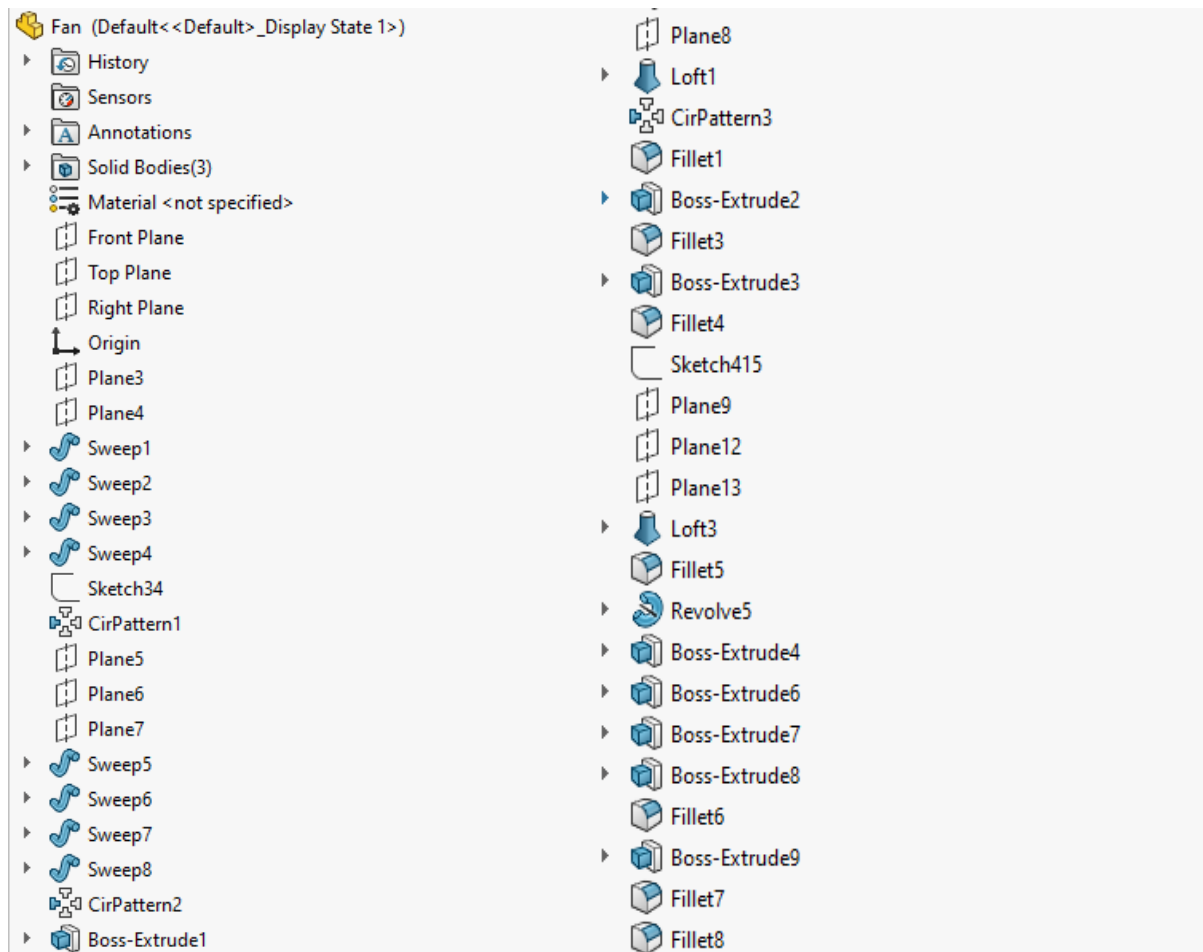
The design focuses on showcasing the structural elements of a wall-mounted fan, including the base, motor housing, blades, and protective grill. This project illustrates the importance of CAD software like SolidWorks in prototyping and simulating real-world products with precision and efficiency.

SOFTWARE TOOL:

Solid Works 2020

FLOW CHART:

The design and modeling process is illustrated.



WORKING:

The project involves designing and assembling a wall-mounted fan model using SolidWorks.

The key steps include:

- **Conceptual Design:** A preliminary sketch of the fan was created, considering aesthetics, functionality, and manufacturing feasibility.
- **Component Modeling:** Each part of the fan was designed separately. The protective grill and blades were modeled with a focus on aerodynamics and safety.
- **Assembly:** The individual components were assembled in SolidWorks using mates to simulate the complete product.
- **Simulation:** Stress and airflow analysis were conducted to evaluate the design's performance under operating conditions.
- **Final Render:** A realistic rendering of the fan was created to visualize the end product.

RESULT:

The project successfully demonstrated the use of SolidWorks for designing a wall-mounted fan. The final model includes:

- A realistic and functional design with detailed components.
- A protective grill ensuring safety while maintaining airflow efficiency.
- An aerodynamic blade design for optimal performance.

Discussion:

The project highlights the importance of 3D modeling tools in prototyping and manufacturing. SolidWorks provided an intuitive platform for creating a detailed model, while its simulation tools helped validate the design before production. Future work can include adding electrical components and testing motor efficiency for a complete functional prototype.



